

Commutator Contribution V3 Summary

This general-loop report keeps the v2 canonical external-label assignment and adds stable multi-loop graph profiles. Loop order: 2. Parsed topology count: 42.

Family histogram

ext:2-1-1—bridges:1-1—loops:1-1: 6. ext:2-1-1—bridges:2-1-1—loops:0: 6. ext:2-1-1—bridges:2-1—loops:1: 12. ext:2-2—bridges:1-1-1—loops:1: 3. ext:2-2—bridges:2-2—loops:0: 3. ext:3-1—bridges:1-1—loops:1-1: 4. ext:3-1—bridges:2-1—loops:1: 4. ext:3-1—bridges:3-1—loops:0: 4.

Structural ladder candidates

Structural ladder-candidate indices: [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 35, 38]. The ladder-candidate flag is a broad structural heuristic: exact for the v2 ‘L=1’ split, and intentionally broader for higher loops where v3 reports graph families rather than exact paper coefficients.

Topology 1

FeynArts topology id: 2. Structural family: ext:2-1-1—bridges:2-1-1—loops:0. Profile: attachments=[[0, 2], [1, 0], [1, 0]], external-multiplicities=[2, 1, 1], self-loops=[], bridges=[2, 1, 1], tree-skeleton=False, ladder-candidate=False.

Topology 2

FeynArts topology id: 2. Structural family: ext:2-1-1—bridges:2-1-1—loops:0. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[], bridges=[2, 1, 1], tree-skeleton=False, ladder-candidate=False.

Topology 3

FeynArts topology id: 2. Structural family: ext:2-1-1—bridges:2-1-1—loops:0. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[], bridges=[2, 1, 1], tree-skeleton=False, ladder-candidate=False.

Topology 4

FeynArts topology id: 2. Structural family: ext:2-1-1—bridges:2-1-1—loops:0. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[], bridges=[2, 1, 1], tree-skeleton=False, ladder-candidate=False.

Topology 5

FeynArts topology id: 2. Structural family: ext:2-1-1—bridges:2-1-1—loops:0. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[], bridges=[2, 1, 1], tree-skeleton=False, ladder-candidate=False.

Topology 6

FeynArts topology id: 2. Structural family: ext:2-1-1—bridges:2-1-1—loops:0. Profile: attachments=[[0, 1], [0, 1], [2, 0]], external-multiplicities=[2, 1, 1], self-loops=[], bridges=[2, 1, 1], tree-skeleton=False, ladder-candidate=False.

Topology 7

FeynArts topology id: 2. Structural family: ext:2-2—bridges:1-1-1—loops:1. Profile: attachments=[[0, 0], [0, 2], [2, 0]], external-multiplicities=[2, 2], self-loops=[1], bridges=[1, 1, 1], tree-skeleton=False, ladder-candidate=False.

Topology 8

FeynArts topology id: 2. Structural family: ext:2-2—bridges:1-1-1—loops:1. Profile: attachments=[[0, 0], [1, 1], [1, 1]], external-multiplicities=[2, 2], self-loops=[1], bridges=[1, 1, 1], tree-skeleton=False, ladder-candidate=False.

Topology 9

FeynArts topology id: 2. Structural family: ext:2-2—bridges:1-1-1—loops:1. Profile: attachments=[[0, 0], [1, 1], [1, 1]], external-multiplicities=[2, 2], self-loops=[1], bridges=[1, 1, 1], tree-skeleton=False, ladder-candidate=False.

Topology 10

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:2-1—loops:1. Profile: attachments=[[0, 2], [1, 0], [1, 0]], external-multiplicities=[2, 1, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 11

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:2-1—loops:1. Profile: attachments=[[0, 2], [1, 0], [1, 0]], external-multiplicities=[2, 1, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 12

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:1-1—loops:1-1. Profile: attachments=[[0, 2], [1, 0], [1, 0]], external-multiplicities=[2, 1, 1], self-loops=[1, 1], bridges=[1, 1], tree-skeleton=True, ladder-candidate=True.

Topology 13

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:2-1—loops:1. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 14

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:2-1—loops:1. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 15

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:1-1—loops:1-1. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[1, 1], bridges=[1, 1], tree-skeleton=True, ladder-candidate=True.

Topology 16

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:2-1—loops:1. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 17

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:2-1—loops:1. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 18

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:1-1—loops:1-1. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[1, 1], bridges=[1, 1], tree-skeleton=True, ladder-candidate=True.

Topology 19

FeynArts topology id: 4. Structural family: ext:3-1—bridges:1-1—loops:1-1. Profile: attachments=[[0, 0], [1, 0], [1, 2]], external-multiplicities=[3, 1], self-loops=[1, 1], bridges=[1, 1], tree-skeleton=True, ladder-candidate=True.

Topology 20

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:2-1—loops:1. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 21

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:2-1—loops:1. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 22

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:1-1—loops:1-1. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[1, 1], bridges=[1, 1], tree-skeleton=True, ladder-candidate=True.

Topology 23

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:2-1—loops:1. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 24

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:2-1—loops:1. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 25

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:1-1—loops:1-1. Profile: attachments=[[0, 1], [1, 0], [1, 1]], external-multiplicities=[2, 1, 1], self-loops=[1, 1], bridges=[1, 1], tree-skeleton=True, ladder-candidate=True.

Topology 26

FeynArts topology id: 4. Structural family: ext:3-1—bridges:1-1—loops:1-1. Profile: attachments=[[0, 0], [1, 0], [1, 2]], external-multiplicities=[3, 1], self-loops=[1, 1], bridges=[1, 1], tree-skeleton=True, ladder-candidate=True.

Topology 27

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:2-1—loops:1. Profile: attachments=[[0, 1], [0, 1], [2, 0]], external-multiplicities=[2, 1, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 28

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:2-1—loops:1. Profile: attachments=[[0, 1], [0, 1], [2, 0]], external-multiplicities=[2, 1, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 29

FeynArts topology id: 4. Structural family: ext:2-1-1—bridges:1-1—loops:1-1. Profile: attachments=[[0, 1], [0, 1], [2, 0]], external-multiplicities=[2, 1, 1], self-loops=[1, 1], bridges=[1, 1], tree-skeleton=True, ladder-candidate=True.

Topology 30

FeynArts topology id: 4. Structural family: ext:3-1—bridges:1-1—loops:1-1. Profile: attachments=[[0, 0], [0, 1], [2, 1]], external-multiplicities=[3, 1], self-loops=[1, 1], bridges=[1, 1], tree-skeleton=True, ladder-candidate=True.

Topology 31

FeynArts topology id: 4. Structural family: ext:3-1—bridges:1-1—loops:1-1. Profile: attachments=[[0, 0], [0, 1], [2, 1]], external-multiplicities=[3, 1], self-loops=[1, 1], bridges=[1, 1], tree-skeleton=True, ladder-candidate=True.

Topology 32

FeynArts topology id: 4. Structural family: ext:3-1—bridges:2-1—loops:1. Profile: attachments=[[0, 0], [0, 1], [2, 1]], external-multiplicities=[3, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 33

FeynArts topology id: 4. Structural family: ext:3-1—bridges:2-1—loops:1. Profile: attachments=[[0, 0], [0, 1], [2, 1]], external-multiplicities=[3, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 34

FeynArts topology id: 4. Structural family: ext:2-2—bridges:2-2—loops:0. Profile: attachments=[[0, 0], [0, 2], [2, 0]], external-multiplicities=[2, 2], self-loops=[], bridges=[2, 2], tree-skeleton=True, ladder-candidate=False.

Topology 35

FeynArts topology id: 4. Structural family: ext:3-1—bridges:2-1—loops:1. Profile: attachments=[[0, 0], [1, 0], [1, 2]], external-multiplicities=[3, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 36

FeynArts topology id: 4. Structural family: ext:2-2—bridges:2-2—loops:0. Profile: attachments=[[0, 0], [1, 1], [1, 1]], external-multiplicities=[2, 2], self-loops=[], bridges=[2, 2], tree-skeleton=True, ladder-candidate=False.

Topology 37

FeynArts topology id: 4. Structural family: ext:2-2—bridges:2-2—loops:0. Profile: attachments=[[0, 0], [1, 1], [1, 1]], external-multiplicities=[2, 2], self-loops=[], bridges=[2, 2], tree-skeleton=True, ladder-candidate=False.

Topology 38

FeynArts topology id: 4. Structural family: ext:3-1—bridges:2-1—loops:1. Profile: attachments=[[0, 0], [1, 0], [1, 2]], external-multiplicities=[3, 1], self-loops=[1], bridges=[2, 1], tree-skeleton=True, ladder-candidate=True.

Topology 39

FeynArts topology id: 6. Structural family: ext:3-1—bridges:3-1—loops:0. Profile: attachments=[[0, 0], [1, 0], [1, 2]], external-multiplicities=[3, 1], self-loops=[], bridges=[3, 1], tree-skeleton=True, ladder-candidate=False.

Topology 40

FeynArts topology id: 6. Structural family: ext:3-1—bridges:3-1—loops:0. Profile: attachments=[[0, 0], [1, 0], [1, 2]], external-multiplicities=[3, 1], self-loops=[], bridges=[3, 1], tree-skeleton=True, ladder-candidate=False.

Topology 41

FeynArts topology id: 6. Structural family: ext:3-1—bridges:3-1—loops:0. Profile: attachments=[[0, 0], [0, 1], [2, 1]], external-multiplicities=[3, 1], self-loops=[], bridges=[3, 1], tree-skeleton=True, ladder-candidate=False.

Topology 42

FeynArts topology id: 6. Structural family: ext:3-1—bridges:3-1—loops:0. Profile: attachments=[[0, 0], [0, 1], [2, 1]], external-multiplicities=[3, 1], self-loops=[], bridges=[3, 1], tree-skeleton=True, ladder-candidate=False.